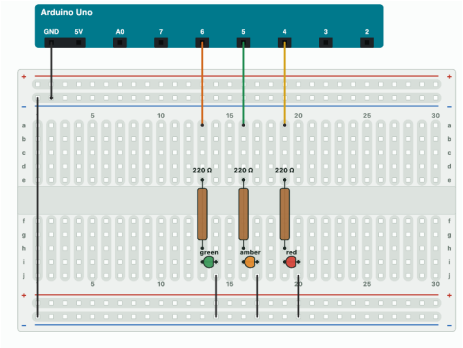
Practical 3 — Test, Amend, Re-test

Booklet 4b, page 21 (Assignment link). Success criterion 4.6. Answers go in your jotter.

The idea being tested

One test for each job the system does. Write what each part did, not 'it didn't work'. Change one thing, then test again.

1 Your traffic lights (Worked Example 2)



Red 4, amber 5, green 6.

Jobs: red, then red and amber, then green, then amber, then back to red. Each stage lasts as long as its delay.

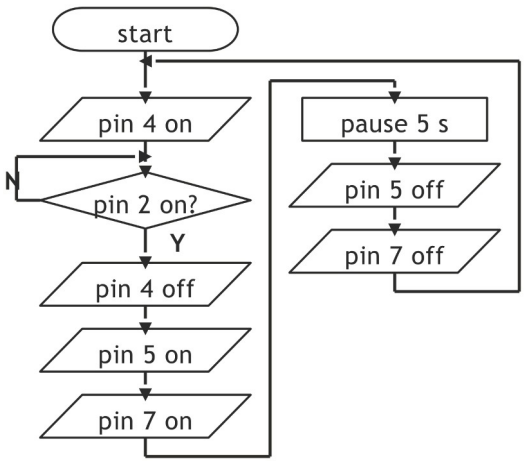
Table 1 — traffic lights

Test	What should happen	What happened	Change planned	Re-test
1				
2				
3				

2 The faulty crossing (NoSyntaxSim, then Yenka)

Table 2 — the crossing's pins

Pin	2	4	5	7
Part	button (in)	red (out)	green (out)	buzzer (out)



Jobs:

1 Red until the button is pressed.
2 Green and beeper for 5 s.
3 Back to red, beeper off.

The file L4_crossing_FAULTY does not match this plan. Your tests find where.

Table 3 — the crossing

Test	What should happen	What happened	Change planned	Re-test
1				
2				
3				

Questions

Q1 Explain why 'it didn't work' is a weak entry in a test table. **[2]**

Q2 Explain why you change only one thing before you test again. **[1]**

Q3 State the fault you found in the crossing, and the change that fixed it. **[2]**

Q4 Suggest one improvement to the crossing. Justify it. **[2]**